

5. The diagram shows an electric hand drier connected to a 230 V a.c. mains supply.



- (a) Complete the following sentence by underlining the correct phrase. [1]

An a.c. or alternating current (**continuously changes direction / continuously changes frequency / continuously changes speed**).

- (b) The electric hand drier has a label attached to it.

Electric hand drier

Power = 2600 W

230 V a.c

- (i) Each time the hand drier is used it blows out hot air for 15 seconds. Use an equation from page 2 to calculate the energy transferred by the hand drier if it is used **6 times**. [3]

Energy transferred = J

- (ii) Use the equation:

$$\text{current} = \frac{\text{power}}{\text{voltage}}$$

to calculate the current when the hand drier is used.

[2]

Current = A



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- (iii) The hand drier is connected to the mains with a standard three-core cable. Complete the table below with the value of the current in each wire in the connecting cable when the hand drier is operating normally. [2]

Wire	Current (A)
live	
neutral	
earth	

- (c) A fault develops and the live wire touches the metal case of the hand drier. Explain how a residual current circuit breaker (rccb) protects the user. [2]

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- (d) A salesperson for a replacement hand drier claims it will use fewer units of electricity each time it is used because its power is 1 300W. When it operates it blows hot air for double the time. Explain if you agree with the claim of the salesperson. [2]

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